CHAPTER 1

Methanol Production and Applications: An Overview

Francesco Dalena*, Alessandro Senatore*, Alessia Marino*, Amalia Gordano†, Marco Basile*, Angelo Basile†

**University of Calabria, Rende, Italy* †*Institute on Membrane Technology (ITM-CNR), Rende, Italy*

Acronyms

ACL	anode catalyst layer
ATR	autothermal reforming
BASF	Badische Anilin und Soda Fabrik
CCL	cathode catalyst layer
CMR	catalytic membrane reactor
DME	dimethyl ether
DMFC	direct methanol fuel cell
ETP	energy technology perspectives
GHG	greenhouse gases
IGCC	integrated gasification combined cycle
MD	methanol decomposition
MSR	methanol steam reforming
NCCC	National Carbon Capture Center
PEM	proton exchange membrane
PO	partial oxidation
POM	partial oxidation of methanol
SR	steam reforming
SW	solid wastes
WGS	water-gas shift
WTE	waste-to-energy

1 Introduction

In the last century, fossil fuels and natural gases have been our major source of energy. Unfortunately, these resources are not renewable and therefore limited. This creates instability in the global market, which leads to a corresponding instability in fuel price. Furthermore,

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fossil fuels are primarily responsible for the emission of greenhouse gases (GHG) such as CO_2, CH_4, and N_2O, which contribute to global warming.

Nowadays, the main competitors appear to be hydrogen and methanol (as discussed in Chapter 25). The use of hydrogen appears to be the most promising from an energy point of view. In fact, it has the highest energy content per unit weight of any known fuel (142 kJ/g), and in comparison to the other known natural gases, it is environmentally safe. At present, the problems related to the use of hydrogen as a new energy resource are the costs of purification processes and the difficulties linked to infrastructure for storage and transport. The most important competitor appears to be methanol, which has an octane number of 113 and a density that is about half that of gasoline (Olah et al., 2009).

Among several uses, methanol can be mixed with conventional gasoline without requiring any technical modification in the vehicle fleet. Most of the methanol-fueled vehicles use M85, a mixture of 85% methanol and 15% unleaded gasoline (Cifre and Badr, 2007). Furthermore, methanol can be used as a convenient energy carrier for hydrogen storage and transportation, as an easily transportable fuel, and also in the chemical industry as a solvent and as a C1 building block for producing intermediates and synthetic hydrocarbons, including polymers and single-cell proteins (Bozzano and Manenti, 2016). For all these reasons, methanol is considered to be the transition molecule from fossil fuels to renewable energies.

Methanol, or methyl alcohol, is an organic compound, CH_3OH, with a molecular weight equal to 32.042 u.m.a. It is only slightly soluble in fat and oil and represents one of the most important chemical raw materials. Indeed, the primary use of methanol is in the chemical industry, as either a feedstock, solvent, or cosolvent.

Approximately 65% of the methanol produced worldwide is consumed for the production of acetic acid, methyl and vinyl acetates, methyl methacrylate, methylamines, methyl ter buthyl ether (MTBE), fuel additives, and other chemicals. The remaining portion is converted into formaldehyde and the resulting products, as shown in Fig. 1.1 (Khadzhiev et al., 2016; Ali et al., 2015).

Many technologies have been developed over the years to produce methanol from different sources. In fact, it can be synthesized from several carbon-containing feedstocks, including natural gases (it could even be produced right at the gas well by oxidative transformation), coal, biomass, or CO_2, the latter directly recovered from the atmosphere.

It is difficult to date precisely the first synthesis of methanol in history. We have tried to give a historical reconstruction of the methanol production processes afterward (Fig. 1.2).

Before the development of the modern industrial era, methanol, also called *wood spirit*, was prepared by wood heating in anaerobic condition. The wood distillation process provided an extract that contained many impurities besides the methyl alcohol. The process was

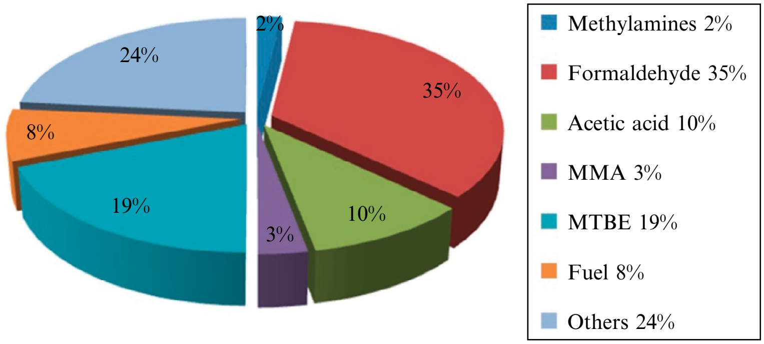

Fig. 1.1
Resulting products of methanol.

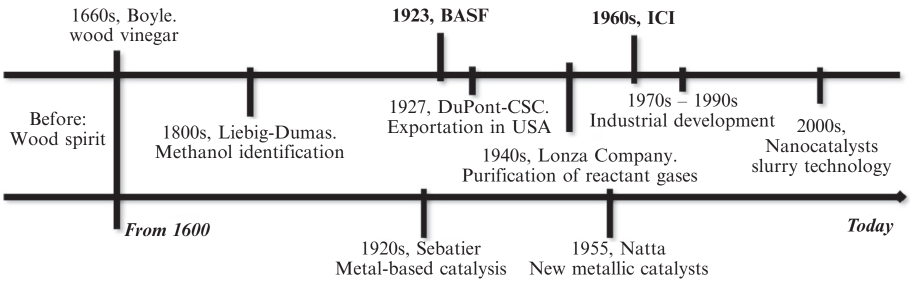

Fig. 1.2
Historical reconstruction of methanol production.

improved in the 1660s by scientist Sir Robert Boyle. He carried out a purification of the extract, called wood vinegar, by a reaction with milk of lime (an aqueous suspension of calcium hydroxide). However, this was not marketed for about two centuries (Fiedler et al., 2000). The exact composition of the wood vinegar remained unknown until the 1800s, when J.V. Liebig and J.B.A. Dumas independently identified the methanol molecule. On the basis of their works, in 1835 the term methyl was formally introduced into chemistry. In the same period, methanol began to be commercialized. A fundamental development in methanol synthesis came from the studies of Paul Sabatier. He found an important way to hydrogenate a large variety of functional groups by metal-based catalysis, and among the numerous compounds he studied, a nickel-based catalyst allowed him to obtain methyl alcohol by hydrogenation of carbon monoxide (Sabatier, 1926). However, the first break in methanol synthesis came in 1923 from a German company called Badische Anilin und Soda Fabrik (BASF). That company developed a metal-based catalytic hydrogenation at high pressure, called the BASF process (Tijm et al., 2001). This technology began to be exported, and in 1927 it was introduced by both DuPont and the Commercial Solvents Corporation in the United States. The BASF process not only represented the starting point in the industrial production of methanol, but it remained the dominant technology for more than 45 years. Over the following years, a great effort has been made to develop new technologies to reduce pressure and temperature levels, with the aim of improving the economic process. In the 1940s, the Swiss Lonza Company began the industrial synthesis of methanol from electrolytic hydrogen and carbon dioxide, the latter derived from $Ca(NO_3)_2$ synthesis. For the first time, the reactant gases had been purified from nitrous vapors, and the reaction had been carried out using the ZnO-based catalyst, developed in Italy by Prof. Giulio Natta, for the methanol synthesis from CO and H_2 (Natta, 1955).

Thanks to the invention of steam methane reforming, which allowed the production of more pure *syngas* (a mixture of H_2, CO, and CO_2), a more active Cu/ZnO catalyst could be used, thus decreasing the process temperature and pressure to about 300°C and 100 bar, respectively. This significant improvement was proposed in the Imperial Chemical Industries (ICI) process in 1966 (Bozzano and Manenti, 2016). On this basis, many implementations were carried out through time and many new industrial plants were adapted to the singular production needs, such as the Lurgi, Haldor-Topsøe, and Linde plants.

Because of the increasing industrial production of methanol, it began to be used in several fields. In 1973, an oil embargo proclaimed by the Organization of Petroleum Exporting Countries (OPEC) in the U.S. and the Netherlands increased the interest in methanol as a new alternative automobile fuel. But while these studies have continued, the U.S. Clean Air Act was passed in 1990. This sanctioned the reduction of ozone and carbon monoxide emissions, which are the methanol combustion products. The use of methanol as a fuel was therefore banned. During the 1990s, market demand grew again when it was discovered that methyl alcohol can also be used as a fertilizer in agriculture for a great variety of products. Today, the global consumption of methyl alcohol is about 92Mt/year (Fig. 1.3), and since 1975 its production has grown at about 1433% (Zhen and Wang, 2015; Khadzhiev et al., 2016).

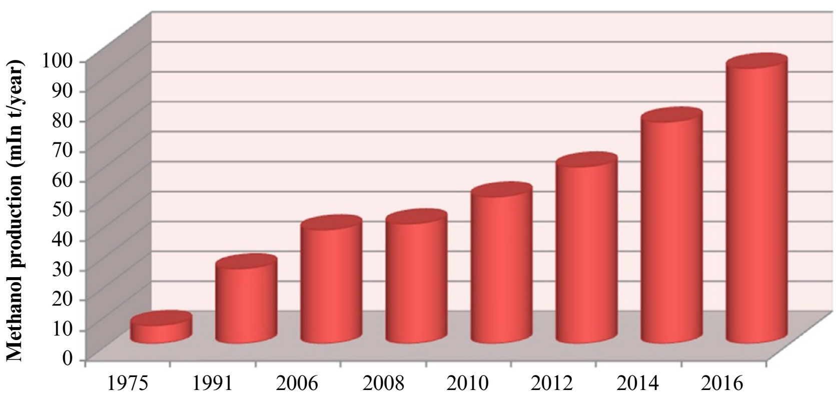

Fig. 1.3
Methanol production expressed in mln ton per year of production from 1975 to 2016.

This chapter seeks to give a general overview of methanol production, its historical production processes (BASF and ICI), and its most innovative trends with a particular focus on the production of dimethyl ether (DME), the production of hydrogen, and the application of methanol in the fuel cells.

2 Methanol Production

As mentioned above, methanol can be produced from several carbon-containing feedstocks, such as natural gas, coal, biomass, and CO_2. For each of them, the reactions and technologies involved are examined in detail.

2.1 Methanol From Natural Gas

Currently, about 90% of methanol is produced from natural gas (Blug et al., 2014). The process route for the production of the simplest alcohol is relatively straightforward, involving the three following basic steps (Tijm et al., 2001; Roan et al., 2004; Fiedler et al., 2005; Spath and Dayton, 2003):

- Production of synthesis gas;
- Conversion of the syngas into crude methanol;
- Distillation of the reactor effluent (crude methanol) to achieve the desired purity.

The mixture of syngas (H_2, CO, and CO_2) is mainly produced by steam reforming (SR) and autothermal reforming (ATR) of natural gas, as shown in Eqs. (1.1) and (1.2), respectively. However, it is also obtained by partial oxidation (PO) of methane (Eq. 1.4) or different carbon-based materials such as coal, heavy oils, or biogas (Wilkinson et al., 2016; Basile et al., 2015).

Steam reforming:

$$CH_4 + H_2O \rightleftarrows CO + 3H_2 \tag{1.1}$$

Autothermal reforming:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \tag{1.2}$$

which involves Eq. (1.1) and a *water gas shift (WGS)* reaction:

$$CO_2 + H_2 \rightleftarrows CO + H_2O \tag{1.3}$$

Partial oxidation:

$$CH_4 + {}^1/_2O_2 \rightarrow CO + 2H_2 \tag{1.4}$$

As reported by Bozzano and Manenti (2016), the composition of syngas is usually characterized by the stoichiometric number S, given by the ratio of two quantities: the difference between hydrogen and carbon dioxide moles, and the summation of the moles of CO_2 and CO.

$$S = \frac{\text{moles } H_2 - \text{moles } CO_2}{\text{moles } CO_2 + \text{moles } CO} \tag{1.5}$$

For the production of methanol and under ideal conditions, S should assume a value of 2. The stoichiometric value S takes into account the presence of CO_2 converted that consumes hydrogen via the reverse WGS reaction (Eq. 1.6).

$$CO_2 + 3H_2 \rightleftarrows CH_3OH + H_2O \quad \Delta H = -41.17\,\text{kJ/mol} \tag{1.6}$$

The value of S depends on the adopted raw material. When syngas is produced by means of natural gas reforming, an S value of 2.8–3 is usually achieved.

The process of converting syngas into crude methanol occurs at a pressure of 50–100 bar and a temperature of 200–300°C. The main reactions involved in methanol synthesis are shown below (Klier, 1982):

Hydrogenation of carbon monoxide:

$$CO + 2H_2 \rightleftarrows CH_3OH \tag{1.7}$$

Divided into two consecutive steps:

$$CO + H_2 \rightleftarrows CH_2O \tag{1.8}$$

$$CH_2O + H_2 \rightleftarrows CH_3OH \tag{1.9}$$

Hydrogenation of carbon dioxide:

$$CO_2 + 3H_2 \rightleftarrows CH_3OH + H_2O \tag{1.10}$$

In which the first step involves the conversion of carbon dioxide in carbon monoxide as expressed in Eq. (1.2).

The reactions (1.7) and (1.10) are strongly exothermic (respectively $\Delta H = -100.46$ kJ/mol in Eq. (1.9) and $\Delta H = -61.59$ kJ/mol in Eq. 1.12) and, consequently, require significant cooling. The mixed gases are fed to the converter with H_2/CO maintained at a ratio from 3:1 to 5:1 for the conventional gas-phase process, which often requires equipment where the WGS is used to boost the hydrogen content (Eq. 1.5). The liquid-phase methanol process, through its superior heat management capabilities, can handle the synthesis gas straight from the generator, as it has a ratio of 1:1 to 1:2 as generated by coal gasifiers (Tijm et al., 2001). However, the ideal ratio H_2/CO_2 responds to Eq. (1.5).

In addition, the conversion of synthesis gas is subjected to a thermodynamic equilibrium that limits the process to a low conversion per pass and, therefore, implies a large recycle of unconverted gas. The resulting recycle and cooling duty are largely responsible for the investment costs of this process segment.

Over the years, several solid catalysts have been developed in order to maximize methanol yield and selectivity and minimize byproduct formation.

2.1.1 The BASF process—high-pressure method

On Jan. 16, 1923, while A. Mittasch and M. Pier were studying ammonia metal-catalyzed synthesis, (the Haber-Bosch process (Tour, 1920)), they discovered that the hydrogenation of carbon monoxide with an iron-based catalyst, over 500°C and 100 bar, provided methanol, among other products. They tried to synthesize methanol, therefore, by waste gases derived from the ammonia synthesis and the same iron-based catalyst. However, this reaction gives different products at the same time (Table 1.1), in which the methyl alcohol is the least thermodynamically favorite (Natta, 1955; Fiedler et al., 2000).

Furthermore, the yield was very low because of pollutants in the reactant gases such as chlorine, hydrogen sulphide, methane, and other hydrocarbons. This caused the deactivation of the iron catalyst and the reaction between carbon monoxide and iron, which can form penta-coordinated complexes (Natta, 1955). Since that time, a great variety of oxides and metals have been tested as hydrogenation catalysts, excluding only the eighth group of the periodic system. All reactions were carried out at high pressure, 250–300 bar, and high temperatures, 320–450°C. Among all, the two catalysts that gave the best results in these reaction conditions were ZnO/Cr_2O_3 and ZnO/CuO (Mittasch et al., 1925; Mittasch and Pier, 1926; Kung, 1980; Chinchen et al., 1988; Tijm et al., 2001; Fiedler et al., 2000).
On this basis, many studies have been done to investigate the reaction mechanisms of the Eqs. (1.6)–(1.8), in presence of the above-mentioned solid catalysts (heterogeneous catalysis). As shown below, two different ways were proposed, both involving the adsorption of CO and H_2 (Kung, 1980): the mechanism A (Fig. 1.4) shows that the reaction takes place in four consecutive hydrogenation steps.

Table 1.1 Byproducts of methanol synthesis

Long-chain alcohols		$nCO + 2nH_2 \rightleftarrows C_nH_{2n+1}OH + (n-1)H_2O$
Which can evolve in	Aldehydes	$RCH_2CH_2OH \rightleftarrows RCH_2CHO + H_2$
	Ketons	$2RCH_2CHO \rightleftarrows RCH_2COCHRCH_3 + O_{ads}$
Hydrocarbons:		$CO + 3H_2 \rightleftarrows CH_4 + H_2O$ $CO_2 + 4H_2 \rightleftarrows CH_4 + 2H_2O$ $nCO + (2n-1)H_2 \rightleftarrows C_nH_{2n+2} + nH_2O$
DME		$2CO + 4H_2 \rightleftarrows CH_3OCH_3 + H_2O$

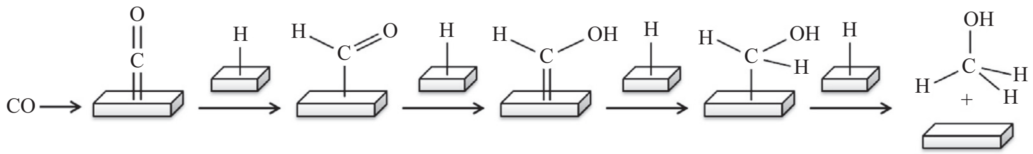

Fig. 1.4
Reaction mechanism A; the gray shape represents the catalyst surface.

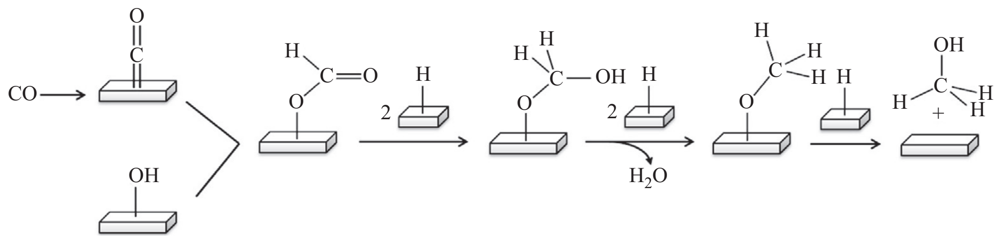

Fig. 1.5
Reaction mechanism B.

In mechanism B (Fig. 1.5) both CO and a hydroxyl group on the catalyst surface are involved. The first step takes place by an insertion of carbon monoxide to form a formate intermediate; subsequent hydrogenation and dehydration lead to the formation of methanol, passing through a methoxide intermediate.

It is noteworthy that the two mechanisms proposed differ not only for the intermediates formed, but also for the way in which they are bonded to the catalyst surface, that is with a carbon atom in mechanism A and with oxygen in B. As a result, catalytic cycles of ZnO/Cr_2O_3 and ZnO/CuO catalysts were proposed. As shown in Fig. 1.6, the active site was assumed to be a cluster of zinc ions with an oxygen vacancy. In accord with these studies, the interaction between the end oxygen of the adsorbed CO and the cluster electron-deficient vacancy allows the activation of the CO bond.

2.1.2 The ICI process—low-pressure method

In the 1960s, the BASF high-pressure method was overcome by a low-pressure method created by ICI (now Johnson Matthey). They developed a new route for methanol synthesis in a pressure range equal to 35–54 bar and at temperatures ranging from 200°C to 300°C. This was made possible not only by the discovery of a new, more active and selective copper-based catalyst ($Cu/ZnO/Al_2O_3$), but also by the development of new advanced purification processes for synthesis gases, which allowed them to use sulfur-chlorine-free syngas (Zhen and Wang, 2015; Van Bennekom et al., 2013).

Despite the fact that the catalytic power of copper/zinc catalysts in methanol synthesis was already well known, this was not exploited commercially due to their low lifetime and low

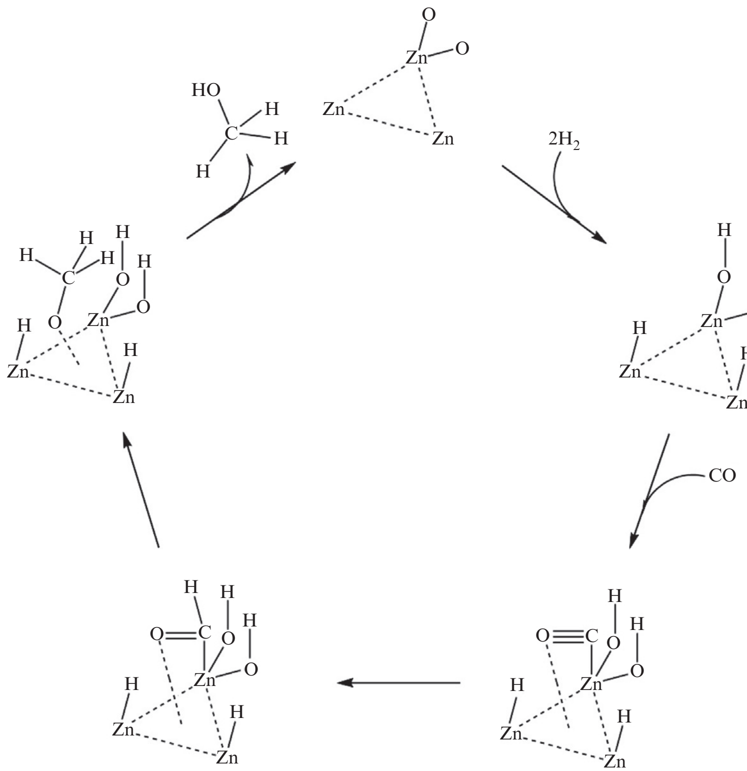

Fig. 1.6
Catalytic cycle of ZnO-catalyzed methanol synthesis.

thermal stability, mostly caused by their deactivation by poisoning (Chinchen et al., 1988; Sá et al., 2010). These problems were overcome thanks to the addition of alumina, which increases the stability of the Cu/ZnO catalyst and inhibits the thermal formation of Cu crystallites (Matsumura and Ishibe, 2009; Sá et al., 2010; Peppley et al., 1999).

Several studies have been carried out in order to investigate the catalytic behavior of the Cu/ZnO/Al_2O_3 catalyst, but today this mechanism is still not fully understood. Recent research confirms what is described above, that is that the oxygen vacancy in the ZnO lattice is responsible for a better adsorption and transformation of CO and CO_2, adding that this presence improves the Cu dispersion on the catalyst support (Ganesh, 2014; Nakamura et al., 2003; Yong et al., 2013).

Furthermore, it is well known that the active site of this kind of catalyst is copper, but there are controversies about its mechanism of action. Many studies suggested that metallic Cu sites are the active catalytic centers. On the other hand, during the methanol synthesis reaction, the migration from ZnO to Cu of the ZnO_x species leads to the formation of oxidized sites Cu^+, also considered as an active catalytic area. Furthermore, it has been shown that both the Cu and Cu^+

species are important in methanol formation, promoting CO_2 and CO hydrogenation respectively, and that the catalytic activity strongly depends on their ratio Cu^+/Cu^0 (Kanai et al, 1994; Fujitani et al., 1994; Ganesh, 2014; Kanai et al. 1996; Nakamura et al., 2003). On this basis, the catalytic cycle shown in Fig. 1.7 has been proposed. The microkinetic model shows that methanol can be produced by hydrogenation of carbon dioxide through the following intermediates HCOO, HCOOH, CH_2COOH, CH_2O, and CH_3O, and can also be produced from carbon monoxide hydrogenation by HCO, CH_2O, and CH_3O intermediates (Grabow and Mavrikakis, 2011).

In Table 1.2 some differences from feedstock and from a production point of view (expressed in t/day) of BASF, ICI, and other some useful production processes for obtaining methanol are reported. As can be seen, the evolution of the processes from the high pressure (BASF) to the low pressure (ICI) led to an increase of 10^5 t/day.

2.2 Methanol From Coal and Biomass

The use of biomass and char represents another route for methanol production. Nowadays, new industrial research is trying to solve the triple problem of energy demand, waste management, and GHG emissions. Waste-to-energy (WTE) technologies convert solid waste into various forms that can be used to supply energy, which can be derived by thermochemical processes. The latter is a better route than biochemical conversion, due to a

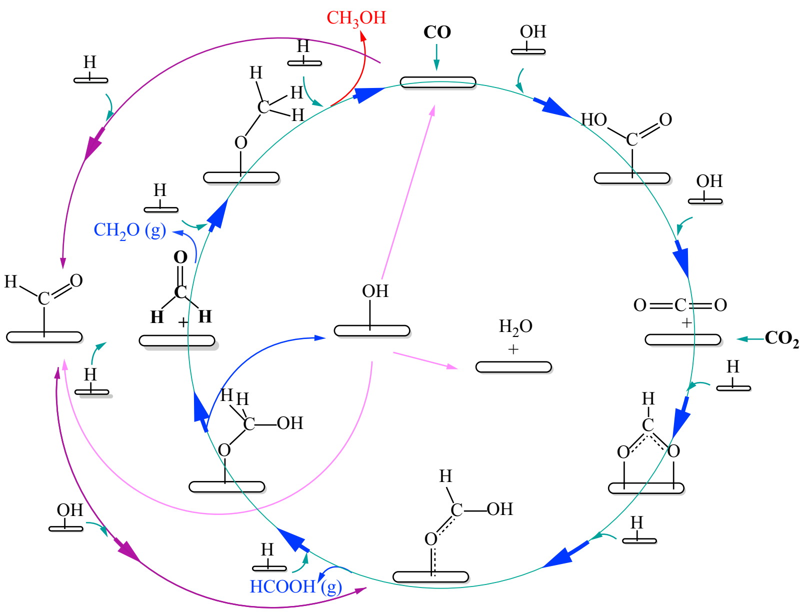

Fig. 1.7
Model of mechanism of methanol synthesis on Cu-based catalyst.

Table 1.2 Methanol yield (t/day) of the main historical industrial processes

Productive process	Feedstock	Yield	Reference
BASF	Syngas	0.07896 t/day	Klier (1982) and Kung (1980), and Mittasch and Pier (1926)
DuPont	Syngas	0.114 t/day	Klier (1982), Kung (1980), and Green (1973)
Haldor-Topsøe	Syngas	2400 t/day	Wilhelm et al. (2001) and Styhr (1972)
ICI	Carbonaceous feedstock	2500 t/day	Pinto (1983)

higher amount of feedstock transformed and a faster conversion rate (Shahbaz et al., 2017; Guan et al., 2016).

The production process for obtaining methanol from coal and biomass is similar to that for the production of methanol from natural gas, subdivided into the following three steps: syngas production, synthesis of the crude methanol, and purification.

In the first step, coal or biomass is converted inside a gasifier into the gaseous products, which consist of biogas (CH_4 and CO_2), syngas (H_2, CO_2 and CO), pure hydrogen, and alkaline gases (Dalena and Basile, 2015).

The gasification process is a thermochemical conversion technique that allows the conversion of solid biomass into gaseous mixtures with the help of gasifying agents such as air/oxygen, steam, and flue gases (Zhao et al., 2015; Dai et al., 2015). This process, using high temperatures in the presence of O_2, aims for a large-scale development to overcome the current limits (i.e., low hydrogen production and high tar content in the syngas (Chen et al., 2016;Shen and Yoshikawa, 2013)). In fact, the unwanted tar may cause the formation of tar aerosols and polymerization into more complicated structures, which are not favorable for hydrogen production through steam reforming (Ni et al., 2006). For this reason, several research groups have focused their studies on maximizing the hydrogen quantity produced and minimizing that of tar.

Under typical gasification conditions, oxygen levels are restricted to less than 30% of that required for complete combustion, and CO and H_2 are the major products (Milne et al., 1998).

Several works in the literature indicate the temperatures necessary to reduce underivatized mono-, di-, and poly-nuclear aromatics (which constitute the tar) to light gases. For example, it has been reported (Corella et al., 1999; Milne et al., 1998) that tar could be thermally cracked at temperature above 1000°C. These authors also carried out their research on biomass gasification under similar conditions but varied the gasifying agent. In fact, the use of some additives such as dolomite inside the gasifier helps tar reduction (Corella et al., 1999).

Generally, dolomite contains 30 wt% CaO, 21 wt% MgO, and 45 wt% CO_2, with others minerals such as SiO_2, Fe_2O_3, and Al_2O_3 (Dalena et al., 2017). It is a cheap disposable catalyst that can significantly reduce the tar content of the product gas from a gasifier.

As reported by Trop et al. (2014), torrefaction (also known as mild pyrolysis), was found to be another productive study aimed at the optimization of biomass properties. This process involves the combustion of the biomass at temperatures in the range of 200–300°C in anaerobic conditions. When the system reaches this temperature, the water and other volatile substances contained within the biomass evaporate, while the lignocellulose (the biopolymer of which is mainly composed the biomass) starts to degrade with a consequent loss of the initial mass equal to 20%. Torrefaction technology greatly reduces the tenacity of biomass; therefore the power consumption needed for grinding torrefied biomass is reduced by 80%–90% in comparison to raw biomass (Van der Stelt et al., 2011). Torrefaction is a promising process for increasing the amount of biomass used for gasification processes. The grindability of torrefied biomass is comparable to that of coal; therefore, it can be gasified in entrained flow gasifiers with high cold gas efficiencies (Trop et al., 2014), but results are environmentally friendly because it has zero carbon footprint.

Methanol is produced from syngas utilizing conventional gasification of biomass at high temperatures (800–1000°C); it can be made with any resource containing carbon, such as coal, or solid wastes (SW) (Balat, 2010).

The conversion of gaseous products involves many reactions described below (Shahbaz et al., 2017; Moghadam et al., 2014; Gao et al., 2012):

Char gasification reaction

$$C + H_2O \rightarrow CO + H_2 \quad \Delta H = 131.5\,kJ/mol \tag{1.10}$$

WGS reaction

$$CO + H_2O \rightleftarrows CO_2 + H_2 \quad \Delta H = -41\,kJ/mol \tag{1.3}$$

Steam reforming of methane

$$CH_4 + H_2O \rightleftarrows CO + 3H_2 \quad \Delta H = 206\,kJ/mol \tag{1.1}$$

Boudouard reaction

$$C + CO_2 \rightarrow 2CO \quad \Delta H = 172\,kJ/mol \tag{1.11}$$

Methanation reaction

$$C + 2H_2 \rightarrow CH_4 \quad \Delta H = -74.8\,kJ/mol \tag{1.12}$$

However, conventional gasification processes applied to biomass do not always produce a syngas with the quality required for methanol synthesis. One of the biggest problems in the gasification of biomass is the formation of tar and char via reduction of carbon oxides, according to Eqs. (1.13) and (1.14):

$$CO + H_2 \rightarrow C + H_2O \quad (1.13)$$

$$CO_2 + 2H_2 \rightarrow C + 2H_2O \quad (1.14)$$

Tar and char formation during biomass gasification can be very problematic when using conventional biomass gasification systems (Shen et al., 2008). However, in order to use biomass in entrained flow gasifiers, the biomass must be pulverized to particle sizes of 100 μm, which is a necessity for these types of gasifiers (Prins et al., 2006; Chen and Kuo, 2010). Large energy consumption is needed to pulverize raw biomass (Trop et al., 2014).

The synthesis gas for methanol production should only contain a small proportion of inert gas components. In fact, the use of air as a gasification agent results in a syngas with a high nitrogen content. For methanol synthesis the optimal H_2:CO_2 molar ratio in the syngas is >2 (Specht and Bandi, 1999) and then the gasification of biomass always results in a gas with a too low H_2:CO_2 ratio. Usually the WGS process is the most frequently used process for ensuring a suitable ratio of CO_2:CO:H_2 and mainly to convert the CO into CO_2. If CO is present when the syngas is combusted in the gasificator, the CO_2 removal efficiency will be limited since the CO in syngas will be converted to CO_2 during the combustion process, as expressed in Eq. (1.3).

The reaction temperature for the WGS is 375°C and the pressure is 40 bar. Carbon monoxide is converted to carbon dioxide and hydrogen, as can be seen in Eq. (1.3). The carbon dioxide produced during the WGS process must be separated from the syngas in order to ensure a suitable ratio of CO_2:CO:H_2 for the commercially available methanol production catalyst required to be 5:28:63 (Zhang et al., 2010; Trop et al., 2014).

To optimize the stoichiometric ratio (Eq. 1.5) and to purify the syngas from substances harmful to the environment, several cleaning processes must be carried out before the gas enters the methanol production process, together with composition optimization. Firstly, sulfur must be removed from the syngas as sulfur is very hazardous for the catalyst used for methanol production and WGS reaction (Trop et al., 2014). Natural gas, in fact, can contain up to 30% hydrogen sulfide (Dalrymple et al., 1991). According to environmental regulations, negligible sulfur content would be allowed in the natural gas and fuels. H_2S, mercaptans, and, in some cases, COS and CS_2 are poisonous sulfur components associated with natural gas. They should be treated via appropriate technologies in gas refineries (Nabikandi and Fatemi, 2015).

In a 2002 report (Korens et al., 2002), SFA Pacific proposed two main methodologies for sulfur purification from commercially available syngas. These methods intend to transform hydrogen sulfide and CO_2 in a mixture of H_2, CO, H_2O, and S_2, according to Eq. (1.14):

$$2H_2S + CO_2 \rightleftarrows H_2 + CO + S_2 + H_2O \quad (1.14)$$

The two main technologies proposed by SPA Pacific are:

- *The sulfinol process*, which is based on a mixture of chemical solvents (aqueous amines) and the physical solvent Sulfolane (tetra-hydrothiophene dioxide). This method has been used in five projects since 1969, including two integrated gasification combined cycle (IGCC) plants and three chemical synthesis applications.
- *The rectisol process*, which is based on refrigerated methanol as the physical solvent. This method continues to be the predominant process used when very pure syngas is required for chemical synthesis. It is normally used where a deep desulfurisation of the synthesis gas is necessary, indicatively for feeds with H_2S composition greater than 30 mol % (in dry basis) (Marcum, 2012). Sulfur compounds are converted to elemental sulfur using a Claus plant and, once purified from sulfur compounds, the syngas is cooled before being introduced into the Rectisol process. The released heat is used to heat the synthesis gas after desulfurisation and to raise low-pressure steam. Part of it is used for the WGS reaction, and the remainder is expanded within a steam turbine. The low-pressure steam is added to the sulfur-free syngas and the mixture is conveyed to the WGS process (Trop et al., 2014).

Another process for the optimization of syngas from a methanol optimization point of view was proposed by the National Carbon Capture Center (NCCC) in 2012 (NCCC, 2012). The NCCC suggests a method to maximize CO_2 capture. In this process, the syngas is passed over a WGS catalyst in the presence of water to convert the CO to CO_2 (Eq. 1.3) prior to the CO_2 removal step. The NCCC is investigating the performance of WGS catalysts under a wide range of steam-to-CO ratios using a WGS reactor incorporated into a gasification power plant. The tests reported in this technical update show that high CO conversions are achieved at lower than recommended steam-to-CO ratios. Furthermore, as molar ratios are increased above about 1.6, the projected improvements in conversions become more marginal. Test data shows that no methane is formed and that no carbon is deposited in the reactor.

Another way to obtain the catalytic conversion of CO to CO_2 is by using a catalytic membrane reactor (CMR). This has possible applications in relation to the recovery of tritium from the blanket of a fusion reactor, to vehicular emission control, to removal of CO from combustion gases, and to conversion of water gas (Basile et al., 1996). By using a CMR, the hydrogen is recovered from the water by a catalytic shift reactor coupled with a membrane permeator. Extracting H_2 from the mixture reaction by using a palladium membrane or a Pd-Ag alloy membrane shifts the reaction toward the products, giving higher conversion with respect to the equilibrium values. The highest CO conversion obtained was 85% at a relatively low temperature (157°C) and at a permeate rate of 0.64 cm^3/min. As mentioned by Basile et al. (1996), it is possible to achieve an almost complete CO conversion (99.9%) at a feed pressure of 1.2 bar and 322°C through a 0.2 μm thick Pd/Al_2O_3 membrane reactor (i.e., considerably mild operating conditions), revealing the dramatic advantage of a membrane reactor over a conventional reactor.

2.3 Methanol From Catalytic Hydrogenation of CO_2

As can be seen in Section 2.1, CO_2 is a linear molecule that is very stable and needs extra efforts to make it reactive. Owing to its high stability ($\Delta G^\circ = -400$ kJ mol^{-1}), a substantial energy input, optimized reaction conditions, and a catalyst with high stability and activity are required for converting CO_2 into value-added chemicals.

As reported by Ganesh (2014), converting one mole of CO_2 to methanol requires an energy input of about 228 kJ and six electrons to reduce C^{4+} of CO_2 to C^{2-} of methanol. The carbon-oxygen bonds are very strong, and high energy is required for breaking them. For this reason, in order to convert CO_2 into methanol, a good catalytic system is required, as expressed in Chapter 15.

The reaction of this catalytic conversion of CO_2 into methanol by hydrogenation is expressed in Eq. (1.10):

$$CO_2 + 3H_2 \rightleftarrows CH_3OH + H_2O \tag{1.10}$$

This method is a useful strategy for CO_2 utilization and a practical approach to sustainable development (Liu et al., 2003). In this conception, CO_2 can be captured from any natural or industrial source, human activities, or air by absorption, and chemically transformed into methanol, then using only renewable resources (*green chemistry*).

Regarding hydrogen, it can originate from water dissociation by electrolysis, as described in Eq. (1.15), using a renewable source of electricity as well (wind, solar, waves, etc.). In this way, it is also possible to store renewable energy on a large, long-term scale (Ganesh, 2014).

$$2H_2O \xrightarrow{\Delta,\ \text{electricity or } h\nu} 2H_2 + O_2 \tag{1.15}$$

Furthermore, other H_2 sources can be biomass pyrolysis or steam/oxygen gasification processes and reforming of biomass-derived products (Bozzano and Manenti, 2016). Other routes are still under investigation. For example, one of them involves the production of biological hydrogen by means of microorganisms activated by sunlight. This biological process of bioenergy conversion of algal biomass into other fuels includes alcoholic fermentation, such as the ABE process, anaerobic digestion, and photobiological hydrogen production. All biological processes are essentially dependent on the presence of enzymes suitable to convert the lignocellulosic biomass in hydrogen (Dalena et al., 2017).

Using CO_2 as a feedstock has many advantages. First of all, it is inexpensive, abundant, nontoxic, noncorrosive, and nonflammable and, therefore, safe to use. Furthermore, it can be easily stored and transported in liquid form under mild pressure and processed in existing syngas conversion plants without any significant modification (Centi and Perathoner, 2009).

According to Olah (2005), methanol production from CO_2 is favorable not only for the use of nonfossil fuel sources (unlike syngas), but also because it allows the avoidance of CO_2 sequestration, which is a very expensive process, and offers mitigation of the greenhouse effect by means of an efficient recycling of CO_2. As reported in *Energy Technology Perspectives 2008* (ETP, 2008), the CO_2 emissions grown in thermal plants are a big problem for GHG emissions. For example, as reported in the review of Ganesh (2014), the CO_2 gas generated at major outlets such as thermal plants and cement industries generates about 4 t of CO_2 for each ton of coal burned. Nowadays, it is therefore necessary to develop a technology that can sequester CO_2 for reuse as a building block for other production cycles. Today, the only technology available to do this from large-scale fossil-fuel usage is CO_2 capture and storage (CCS).

The *Energy Technology Perspectives 2016* (ETP, 2016) scenarios demonstrate that CCS will need to contribute nearly one-fifth of the necessary emissions reductions to reduce global GHG emissions by 70% by 2050 at a reasonable cost. As reported by Tanaka (2008), CCS is therefore essential to the achievement of deep emission cuts. In a scenario that aims at emissions stabilization based on options with costs up to \$50/t CO_2, 27 gigatons (Gt) of CO_2 would need to be captured and stored by 2050, which is 70% of the total emissions. In the ETP scenario, which cuts global CO_2 emissions more than half and which considers emission abatement options with a cost of up to \$200/t CO_2, CCS accounts for 19% of total emissions reductions in 2050.

3 Methanol Application

As discussed in Section 1, the main methanol demand is in the chemical market; about 35% of it is consumed for formaldehyde production. The remaining volumes are consumed for the production of fuel additives, acetic acid, methyl and vinyl acetates, and other chemicals. Recently, methanol synthesis has become the second source of hydrogen consumption after ammonia production (Olah et al., 2011).

3.1 Methanol to DiMethylEther

In the last 10 years, one of the most promising technologies is the use of methanol as a C1 building block in the petrochemical industry, and a wide part of its production is consumed in the manufacturing of DME as an alternative fuel. DME has an octane number and ignition temperature close to that of diesel fuel. It leads to lower NO_X emissions, less smoke, and less engine noise than conventional diesel engines and, furthermore, can be easily transported (Semelsberger et al., 2006; Hosseininejad et al., 2012).

DME can also be used as a chemical feedstock for manufacturing many products, such as short olefins (ethylene and propylene), gasoline, hydrogen, acetic acid, and dimethyl sulfate. The current DME manufacturing process is a double-step (indirect) synthesis, in which methanol synthesis, carried out by one of the above-mentioned technologies, is followed by dehydration, according to Eq. (1.16).

$$2CH_3OH \rightarrow CH_3OCH_3 + H_2O \tag{1.16}$$

Another developing process involves the direct synthesis of DME from syngas. Research on this technology, capable of mass-producing DME at low cost, is underway by JFE Holdings. In this process, syngas CO and H_2, with a 1:1 molar ratio, are converted in DME in the presence of a catalyst. The overall reaction of the process (Eq. 1.17) is shown below (Ohno et al., 2006).

$$3CO + 3H_2 \rightarrow CH_3OCH_3 + CO_2 \tag{1.17}$$

In the late 20th century, the global production of DME was as low as 100–150 kt/year, but it is predicted, for example, that demand just in South Korea will exceed 6.5 Mt by 2020 (Khadzhiev et al., 2016). The production process of this promising fuel is widely discussed in Chapter 11.

3.2 Hydrogen Production

Hydrogen is considered a clean energy source that has an important role in refining, the chemical industry, and the electronic industry. Nevertheless, as seen in the introduction, hydrogen is difficult to store and transport, which seriously restricts its application. Its production from an easily transported liquid feedstock can be an efficient alternative. Methanol is considered an excellent liquid H_2 source with low toxicity and low chain-alcohols (Lü et al., 2012). For this reason, many research groups are developing various technologies that utilize methanol in the production of hydrogen.

These technologies are developing some thermochemical methods: direct decomposition (Lü et al., 2012; Choi and Kang, 2007; Usami et al., 1998), SR reaction (Sá et al., 2010; Yao et al., 2006; Chen and Lin, 2013; Iulianelli et al., 2014; Chen and Syu, 2011), PO (Chen et al., 2015) and also by methanol-water solution electrolysis (Damle, 2008).

3.2.1 Methanol decomposition

Methanol decomposition (MD) is an endothermic reaction to produce H_2 and CO, according to Eq. (1.18):

$$CH_3OH \rightleftarrows CO + 2H_2 \quad \Delta H = 90.1\,\mathrm{kJ\,mol^{-1}} \tag{1.18}$$

This is applicable to the recovery of waste heat of around 200°C from industries (Usami et al., 1998). For the development of heat-recovery systems, new catalysts that can be active even at temperatures below 200°C are indispensable. Usami et al. (1998), for example, have tested 15% wt Pd/ZrO_2, Pd/Pr_2O_3, and Pd/CeO_2 catalysts. These catalysts, prepared by co-precipitation, can produce high catalytic activity in the selective decomposition of methanol to CO and H_2 at a temperature as low as 200°C.

3.2.2 Methanol steam reforming

The production process that is much used in the production of hydrogen is the methanol steam reforming (MSR) (Sá et al., 2010).

Compared to other fuels, methanol presents several advantages for hydrogen production. In fact, the absence of a strong C—C bond facilitates the reforming at low temperatures (200–300°C), a range of temperatures that is very low when compared to other common fuels (800–1000°C for methane and 400°C for ethanol) (Iulianelli et al., 2014).

The overall reaction of the MSR to the production of H_2 is expressed in Eq. (1.19):

$$CH_3OH + H_2O \rightleftarrows CO_2 + 3H_2 \quad \Delta H = 49.7\,\mathrm{kJ\,mol^{-1}} \tag{1.19}$$

In this reaction, two side reactions must be considered: the decomposition of methanol (Eq. 1.20) and the WGS (Eq. 1.3) as expressed below:

$$CH_3OH \rightleftarrows CO + 2H_2 \quad \Delta H = 90.2\,\mathrm{kJ\,mol^{-1}} \tag{1.20}$$

Even though the purpose of the MSR reaction is the production of hydrogen.

As indicated from the values of ΔH in the equations, only the WGS reaction is exothermal ($\Delta H = -41.2$ kJ mol^{-1}) and takes place without the variation of any mole number. The MSR reaction, besides being endothermic, takes place with an increase in the mole number. Unfortunately, the main drawback of this process is represented by the CO formation as a byproduct; this formation can deactivate the catalyst. Nowadays, there are several research groups that pay special attention to catalyst optimization in order to reduce the CO content (Iulianelli et al. 2014).

The steam reforming of methanol, the update of the use of catalysts for this technology, is widely discussed in Chapter 9.

3.2.3 Methanol-water solution electrolysis

Another way to produce hydrogen from methanol is by methanol-water solution electrolysis using an electrolytic cell. Generally, electrolysis of water is the best option for producing very pure hydrogen very quickly (Menia et al., 2017). As in the electrolysis of water, in the methanol-water system the hydrogen produced is very pure (hydrogen concentration is 95.5–97.2 mol%), but the theoretical voltage of the system is much lower than in water electrolysis (1.23 V in water electrolysis compared to 0.03 V in methanol-water solution electrolysis). The full reaction system is well explained by Take et al. (2007).

In methanol-water solution electrolysis, hydrogen is produced by applying DC voltage to the electrolytic cell, as can be seen in Fig. 1.8.

In this system the overall reaction is:

$$CH_3OH + H_2O \rightarrow CO_2 + 3H_2 \tag{1.21}$$

The reaction at the anode and at the cathode are, respectively, Eqs. (1.22) and (1.23):

$$CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^- \quad \text{anode} \tag{1.22}$$

$$6H^+ + 6e^- \rightarrow 3H_2 \quad \text{cathode} \tag{1.23}$$

At the anode, methanol reacts with water to produce carbon dioxide (exhausted outside the anode), protons, and electrons, according to the reaction (1.22).

The protons produced by the anode reaction move to the cathode of the electrolytic cell through the proton exchange membrane (PEM) and the electrons produced by the anode reaction move to the cathode through the external circuit containing the DC power supply.

At the cathode, protons supplied from the anode react with electrons supplied from the anode, according to Eq. (1.23).

Hydrogen production by methanol-water solution electrolysis is suitable for portable power applications because methanol-water solution electrolysis can start up and shut down in a moment and can produce hydrogen at a low temperature. The voltage needed in methanol-water is three times lower than that for water solution electrolysis. Furthermore, the work of Menia et al. (2017) shows that this technique must be improved by changing some parameters,

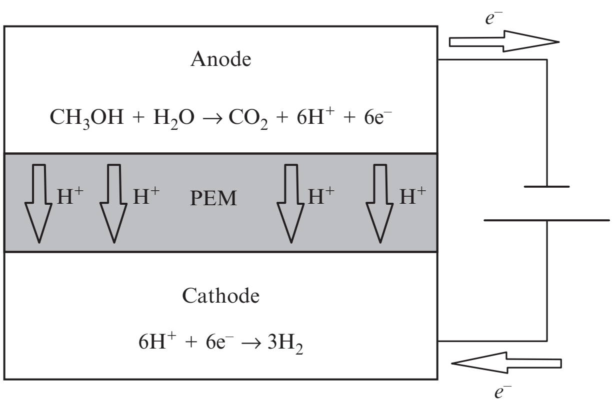

Fig. 1.8
Methanol-water solution electrolysis.

in particular the operating temperature, the cell voltage, the methanol concentration, and the nature of the catalysts.

3.2.4 Partial oxidation/autothermal reforming

Unlike MD and MSR, partial oxidation of methanol (POM) is an exothermic reaction, so no additional heat is needed. However, the temperature control can be difficult. The addition of steam into POM leads to the ATR of methanol. In other words, methanol ATR is a combined reaction of MSR and POM, and its H_2 yield is between the two reactions. The addition of steam in ATR intensifies H_2 production but lowers the reaction temperature and thereby the reaction rate.

On account of the simultaneous addition of oxygen and steam in methanol ATR, the thermal behavior and operating control become more complicated and intractable. In contrast to MSR and methanol ATR, relatively less H_2 is produced from POM.

POM is expressed as:

$$CH_3OH + {}^1/_2O_2 \rightleftarrows 2H_2 + CO_2 \quad \Delta H = 192.3\,\mathrm{kJ\,mol^{-1}} \tag{1.24}$$

Various studies of POM concerning a variety of catalysts are under development in order to optimize the experimental conditions. Normally, the reaction temperature is controlled at 150–300°C.

By virtue of more heat released from POM being employed for chemical reactions, POM is kinetically faster than MSR and methanol ATR, implying that a smaller reactor is achievable for POM.

Instead, ATR (Eq. 1.25) is a combination of MSR and POM. The overall reaction is:

$$4CH_3OH + 3H_2O + {}^1/_2O_2 \rightleftarrows 11H_2 + 4CO_2 \quad \Delta H = -44\,\mathrm{kJ/mol} \tag{1.25}$$

This process uses the energy produced from POM to supply the endothermic MSR reaction, and thus it can be run adiabatically (Kulprathipanja and Falconer, 2004). The heat released from POM can rapidly start up the reaction and drive the endothermic reaction of MSR. The overall reaction can be controlled to be thermo-neutral or slightly exothermic by adjusting the O_2/C molar ratio (Chen and Lin, 2013). This process is discussed in Chapter 9.

3.3 Methanol Fuel Cells

Nowadays, systems that require an external power supply for charging are constantly growing (computers, mp3 players, mobile devices, etc.). For this reason, researchers are looking for viable solutions to systems based on a rechargeable battery of Li and Ni. Methods to implement this technology from an energy point of view and especially to eliminate external sources for charging fuel cells are under study. In particular, PEM fuel cells are electrochemical devices that convert chemical energy into electrical energy. One type of PEMFC is the direct methanol fuel cell (DMFC) that uses methanol or methanol solutions as fuel and works at an ambient temperature (Li and Faghri, 2013). The system is very similar to methanol-water solution electrolysis. The overall reaction is:

$$CH_3OH + 1.5O_2 \rightarrow CO_2 + 2H_2O \tag{1.26}$$

The structure of the DMFC consists of two porous electrocatalytic electrodes on both sides of a solid polymer electrolyte membrane. The thermodynamic reversible potential for the overall cell reaction is 1.214 V (Mallick et al., 2016).

Methanol and water are oxidized in the anode catalyst layer (ACL) and release electrons and protons, according to reaction (1.27):

$$CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^- \tag{1.27}$$

The electrons are transported through an external circuit to the cathode, while the protons penetrate the electrolyte membrane to the cathode. In the cathode catalyst layer (CCL), the oxygen from the ambient atmosphere reacts with electrons and protons and generates water, according to reaction (1.28) (Mallick et al., 2015):

$$6H^+ + 6e^- + 1.5O_2 \rightarrow 3H_2O \tag{1.28}$$

This process and its applications, the use of catalysts, and future trends are widely discussed in Chapter 14.

Conclusions and Future Trends

A comprehensive overview of methanol's applications and production potential has been given in this chapter. This simple alcohol is one of the biggest future frontiers for green chemistry, both in terms of production and application.

From a production point of view, two of the three production cycles of methanol take biomasses or CO_2 as starting feeds. In each case, it's about new technological frontiers that are suitable for reducing GHG emissions. In fact, in the first case, GHGs are reduced by means of WTE technologies, in which solid wastes are converted into various forms that can be used to supply energy. This also helps solve at the same time the problem of waste management. In the second case, it can be possible to effect a direct reduction of the amount of one of the most important GHGs ($CO_{2)}$, which can be captured from any natural or industrial source, human activity, or air by absorption, and chemically transformed into methanol. In addition, it is noteworthy that the production of methanol from CO_2 can be regarded as a completely green process, considering that the hydrogen necessary for this productive cycle can be originated from water dissociation by electrolysis, also using a source of renewable electricity such as wind, solar, waves, etc.

As regards the application point of view, methanol is one of the most innovative and versatile molecules. Nowadays, it is not only used as a solvent and C1 building block for producing intermediates and synthetic hydrocarbons, including polymers and single-cell proteins, but also as an easily transportable fuel and convenient energy carrier for hydrogen storage and transportation.

This chapter, in which attention is paid to only some of the most innovative trends such as the production of DME, the production of hydrogen, and the DMFC, represents a necessary general overview for a better comprehension of this book. The production processes, the applications, the modeling, the most innovative technologies, and the role in the market of this molecule will be discussed in detail in the next chapters.

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